301. A domestic loudspeaker indicates 30W as its maximum continuous power and a sensitivity of 86dB$_{SPL}$/1W at 1m. What will be the maximum radiated pressure at 10m assuming free-field and spherical radiation?

302. We have a loudspeaker radiating a sine wave of 1KHz.

a. If the sensitivity of the loudspeaker is 90dBSPL, and we feed it with 2W, calculate the SPL at 1m.
b. Calculate the Sound Power level for the loudspeaker radiating in free field

303. We have a loudspeaker of $Z_{NOMINAL}$=8Ω that has a sensitivity of 93dB$_{SPL}$ at 1W (2.83V_{RMS}, 1KHz), at 1meter. The manufacturer indicates the following maximum (electrical) powers supported:

- $W_{CONTINUOUS}$ = 120W
- $W_{AVERAGE}$ or $W_{PROGRAM}$ = 240W
- W_{PEAK} = 400W

a. Explain the differences in the application and concept of the three power values.

b. Calculate the electroacoustic performance η[%] of the loudspeaker?

c. Determine, for the three electrical powers, the sound pressure level in dB$_{SPL}$ at 1m and the acoustic power (dB$_{SWL}$ - remember the acoustic power reference, $W_{REF} = 10^{-12} W_{AC}$)

d. Suppose the loudspeaker had a sensitivity 3dB lower, how much should we increase the power to obtain the same sound pressure level?

Imagine that we feed the loudspeaker with an amplifier that can deliver a maximum 50W continuous power over a 8Ω load. We progressively increase the program signal input to the amplifier until its output "clips" severely, converting a sinusoidal input into an output that we will consider almost square wave. Keeping in mind the crest factor for sinusoidal and square waves.

e. What would be the corresponding voltage (V_{RMS}), the continuous power delivered to the loudspeaker and the corresponding sound pressure level value (dB$_{SPL}$)? What would be the possible consequences of this situation both for the quality of the amplified sound and for the integrity of the loudspeaker?

f. Suppose now that the amplifier delivers 100W instead of the 50W of the previous section. What would happen to the loudspeaker?

(based on The Yamaha Sound Reinforcement Handbook, pp. 201-205)

304. A power amplifier has 10kΩ line input impedance. We connect the amplifier to a 4Ω input impedance loudspeaker. If the amplifier receives a +4dBu line level signal,

a. What will be the electrical power in dB_m at the amplifier input?

b. What will be the electrical power in dB_m sent to the loudspeaker when it receives a $4V_{RMS}$ voltage signal?

c. What is the power gain (ratio) in dB?

d. If the loudspeaker sensitivity corresponds to $90dB_{SPL}$ ($1W_{RMS}$, 1m), what would be the sound pressure level measured at 2m from the loudspeaker?

305. Consider two PA loudspeakers in free-field (l_1 and l_2) that are placed with one separation of 1 meter and radiate monophonically the same sound pressure level. As shown in the figure below, a microphone is located on-axis 2 meters away from l_1.

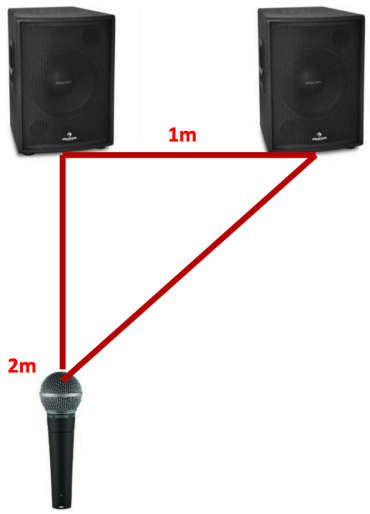
a. What are the first two frequencies to be canceled (destructive interference, f_{D1} and f_{D2}) at the microphone placement?

b. What would be the first frequency where constructive interference (f_{C1}) will appear in the microphone placement?

c. What type of interfering pattern do we get at very low frequency?

d. Explain briefly the effect on the interfering pattern of moving the microphone closer to the bisector between the loudspeakers.

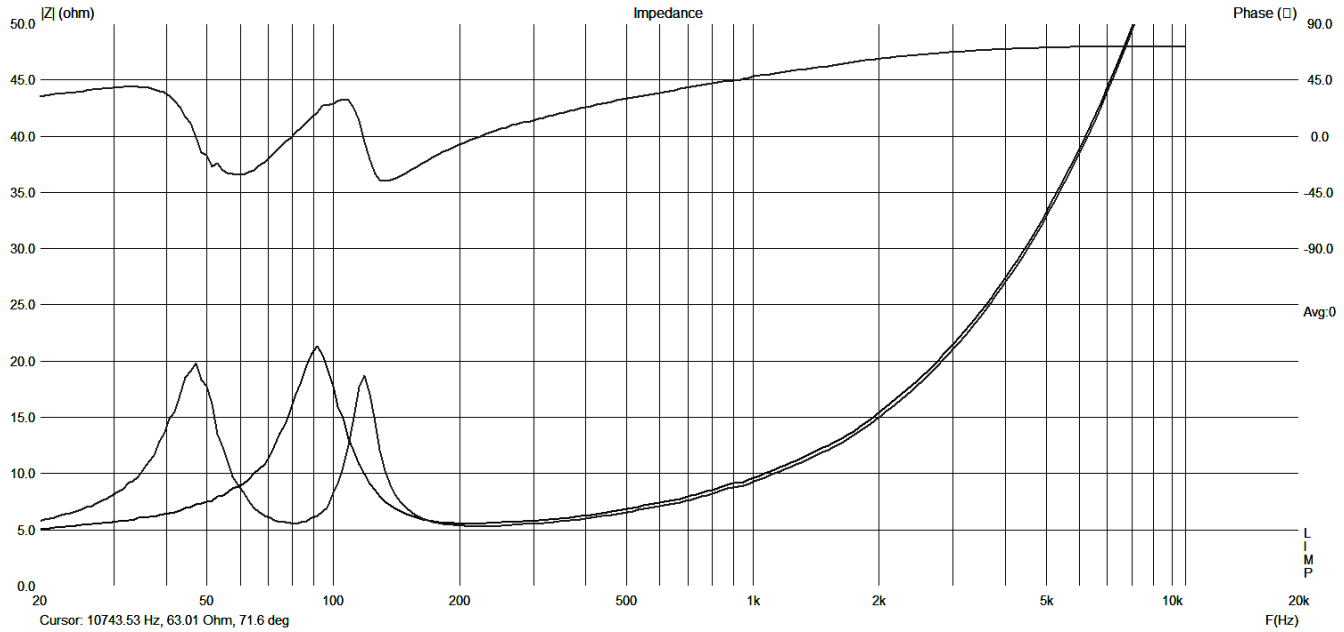
e. Explain how the result might vary if we were located in a room with a lot of diffusion and reverberation phenomena (instead of the free-field assumption).



306. We have four loudspeakers of 8Ω and a mono amplifier of 100W at 8Ω with a minimum impedance of 2Ω.

a. Calculate the equivalent load impedance if we connect the four loudspeakers in parallel. (2,5 points)
b. If a cluster formed by the four loudspeakers in parallel of the previous section generates 120dB SPL at 1m, calculate the theoretical level generated only by one loudspeaker at that distance. (2,5 points)
c. We are asked to add four additional loudspeakers connected also in parallel. Explain briefly if you would recommend this configuration. (2,5 points)
d. Illustrate through a diagram a solution to connect the eight loudspeakers to the amplifier with a total load impedance of 4Ω. (2,5 points)

307. For the next impedance response (magnitude and phase):



a. Justify which magnitude curve corresponds to the sealed box enclosure, and which one corresponds to the bass-reflex.
b. Justify briefly which of the two configurations is represented in the phase response.
c. Indicate all the characteristic values that can be obtained from this phase response.
d. On the magnitude response of the sealed box, indicate the fundamental resonance frequency.
e. On the magnitude response of the bass-reflex configuration, indicate the resonance frequency of the port
f. Attending the bass-reflex magnitude response, describe the relationship between the resonant frequency of the box, and the resonant frequency of the driver.